

Data and text mining

BACPI: a bi-directional attention neural network for compound–protein interaction and binding affinity prediction

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Abstract

Motivation: The identification of compound–protein interactions (CPIs) is an essential step in the process of drug discovery. The experimental determination of CPIs is known for a large amount of funds and time it consumes. Computational model has therefore become a promising and efficient alternative for predicting novel interactions between compounds and proteins on a large scale. Most supervised machine learning prediction models are approached as a binary classification problem, which aim to predict whether there is an interaction between the compound and the protein or not. However, CPI is not a simple binary on–off relationship, but a continuous value reflects how tightly the compound binds to a particular target protein, also called binding affinity.

Results: In this study, we propose an end-to-end neural network model, called BACPI, to predict CPI and binding affinity. We employ graph attention network and convolutional neural network (CNN) to learn the representations of compounds and proteins and develop a bi-directional attention neural network model to integrate the representations. To evaluate the performance of BACPI, we use three CPI datasets and four binding affinity datasets in our experiments. The results show that, when predicting CPIs, BACPI significantly outperforms other available machine learning methods on both balanced and unbalanced datasets. This suggests that the end-to-end neural network model that predicts CPIs directly from low-level representations is more robust than traditional machine learning-based methods. And when predicting binding affinities, BACPI achieves higher performance on large datasets compared to other state-of-the-art deep learning methods. This comparison result suggests that the proposed method with bi-directional attention neural network can capture the important regions of compounds and proteins for binding affinity prediction.

Availability and implementation: Data and source codes are available at <https://github.com/CSUBioGroup/BACPI>

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1 Introduction

The identification of interactions between drug compounds and their potential target proteins plays a critical role in drug discovery and development (Kola and Landis, 2004; Paul *et al.*, 2010). Finding potential compound–protein interaction (CPI) is helpful for a particular protein to choose a specific compound and discover new uses for known drugs. Although high-throughput screening, genomics, proteomics and systems biology have been widely applied for drug development, experimental characterization of compound–protein pair from a large-scale remains to be extremely costly and time-consuming (Broach and Thorner, 1996; Bakheet and Doig, 2009; Chu and Chen, 2008; Rieke *et al.*, 2006). Considering the huge cost of traditional experimental methods, various

computational methods have been proposed to predict potential CPIs. These methods can be divided into two major categories: structure-based methods and structure-free methods.

Structure-based methods can provide insights into action mode about CPIs and predict their binding affinity, such as molecular docking, pharmacophore modeling, chemogenomic relationships and quantitative structure–activity relationship (Cheng *et al.*, 2012; Gilson and Zhou, 2007; Koutsoukas *et al.*, 2011; Trott and Olson, 2010; Xie *et al.*, 2012). These methods provide good biological interpretation, as they can predict potential binding sites on 3D structures of compound–protein complexes. However, these methods have the limitations of extremely computational resources-consuming and over rely on the available 3D structure data.

To eliminate this bottleneck, a variety of structure-free methods including machine learning-based, network-based and matrix factorization-based have been developed for CPI prediction. For example, Yamanishi *et al.* (2008) proposed a supervised learning method based on kernel regression to infer CPIs by integrating the chemical and genomic spaces. Furthermore, Bleakley and Yamanishi (2009) presented a novel supervised learning method of bipartite local model (BLM) that applies support vector machine (SVM) with known interactions and similarity between chemical structures and protein sequences. In order to solve the cold start problem, Mei *et al.* (2013) further presented a new method called BLM-NII integrating Neighbor-based Interaction-profile Inferring (NII) into the existing BLM method, and Van Laarhoven and Marchiori (2013) used regularized least squares and integrated the weighted nearest neighbor into the existing Gaussian interaction profile kernel. Recently, several network-based methods have been developed for CPI prediction by incorporating various biological data, such as chemical, genomic, pharmacological, and phenotypic. Wang *et al.* (2014) proposed a three-layer heterogeneous graph model to capture relationships among diseases, drugs and targets. And Luo *et al.* (2017) presented a novel network integration pipeline for CPI prediction, which integrated diverse information from heterogeneous data sources and learned low-dimensional representations. Considering the importance of drug compound and target protein profile information when building the model, Hao *et al.* (2017) proposed a dual-network integrated logistic matrix factorization algorithm to predict potential CPIs. Although relatively effective of these structure-free methods, they still suffer from several limitations. One major limitation is that most structure-free methods neglect the bioactivity information of binding affinity values about protein-ligand interactions, and mainly focus on a binary classification problem. In addition, without taking advantage of the structural information, their interpretability is still limited.

CPI is not a simple binary on-off relationship (Pahikkala *et al.*, 2015), but a continuous value named binding affinity, which reflects how tightly the compound binds to a particular target protein. The binding affinity between a compound and a target protein is quantified by measures such as dissociation constant (K_d), inhibition constant (K_i) or the half-maximal inhibitory concentration (IC_{50}). The low value of the three measures indicates high strength binding. Prediction of compound-protein binding affinity is a regression-based problem, which aims to predict an approximate value of CPI strength. Recently, several compound-protein binding affinity prediction methods have been presented as a scoring function to replace the virtual screening and molecular docking (Gabel *et al.*, 2014; Li *et al.*, 2015; Shar *et al.*, 2016).

More recently, deep learning-based models have achieved attractive performance in various fields, such as speech recognition, object recognition, recommended system and bioinformatics (Liang and Hu, 2015; Li *et al.*, 2019; Nassif *et al.*, 2019; Wang *et al.*, 2021; Zhang *et al.*, 2020; Zhou *et al.*, 2018), which have been proposed to predict compound-protein binding affinity by employing several deep neural network (DNN) architectures, such as recurrent neural network, convolutional neural network (CNN) and graph convolutional neural network to process and extract contextual features from the input discrete data, which requires only the simplified molecular input line entry system (SMILES) format of compounds and the amino acid sequences of proteins as input (Karimi *et al.*, 2019; Li *et al.*, 2020; Öztürk *et al.*, 2018). Benefiting from the powerful expressive of DNN by non-linear transformations in each layer, and thus these methods can successfully predict the binding affinity.

In this study, we propose an end-to-end neural network model, named BACPI, to predict both CPI and their binding affinity, which directly take low-level representations of compounds and proteins as input and employ graph attention network (GAT) for compounds and CNN for proteins. Specifically, compounds are represented as atoms adjacency graphs, and protein are represented as amino acid sequences. And, we use GAT and CNN to learn the representations of atoms and amino acids, respectively. In addition, we design bi-directional attention (attention of atoms to protein and amino acid to compound) neural network architecture to integrate the

representations of compounds and proteins, which can guide the model to focus on local effective sites of atoms and amino acids, and thus increase the interpretability of the model. Finally, the integrated vector representations of compound-protein pair are fed into the classifier to predict CPIs or regression model to predict compound-protein binding affinity.

In our computational experiments, we use binary interaction datasets and continuous affinity datasets to evaluate the performance of BACPI in CPIs and binding affinities prediction task. Comprehensive cross-validation tests on balanced and unbalanced datasets demonstrate that our model can achieve superior performance in predicting CPIs. This indicates that BACPI with end-to-end representations learning of compounds and proteins is more flexible and robust than traditional machine learning-based methods. And in compound-protein binding affinities prediction task, the experiment results show that BACPI can achieve higher performance on large datasets or comparable performance on small datasets compared to other state-of-the-art deep learning methods. This suggested that the bi-directional attention neural network trained on a large amount of samples can capture the important regions of compounds and proteins for binding affinity prediction. All these results suggested that BACPI can be effectively applied to predict the interaction and the binding affinity of compounds and proteins.

2 Materials and methods

In this section, we describe the details of BACPI. As shown in Figure 1, BACPI model includes a GAT, a CNN and a bi-directional attention neural network. Given the SMILES format of the compound and the amino acid sequence of the protein, we first use RDKit to convert the compound into a graph and split the amino acid sequence into diverse subsequences which are regarded as residues. Then, a GAT and a CNN are used to learn the representations of the atom and residue from the atom structure graph and residue sequence, respectively. Finally, these atom and residue features are fed into the bi-directional attention neural network to integrate the representations and capture the important regions of compounds and proteins, and the integrated features are used to predict the CPI or binding affinity.

2.1 Compound representation learning

The GAT allows for assigning different importances to different nodes within a neighborhood, instead of the mean over all neighbors' representation vectors. And GAT can learn and update representation vectors for every atom of compound by iteratively gathering information from the neighbors of each atom, so that each single atom extracts the local feature of the compound substructures. We first use RDKit to convert the SMILES format of compound to a graph representation. A graph is represented as $G = \{V, E\}$, where V is the set of vertices and each vertex $v_i \in V$ represents i th atom, and E is the set of edges and $e_{ij} \in E$ is the chemical bond between the i th atom and j th atom. Then graph G and the randomly initialized embeddings of vertices are fed into the graph attentional layer. We represent the embeddings of vertices as $\mathbf{v} = \{\vec{v}_1^{init}, \vec{v}_2^{init}, \dots, \vec{v}_{N_v}^{init}\}$, $\vec{v}_i^{init} \in \mathbb{R}^{H_c}$, where N_v is the number of vertices (i.e. atoms) in graph G and H_c is the dimension of vertex embedding. A learnable linear transformation is applied to increase expressive power and transform the embeddings into higher-level features $\mathbf{v}' = \mathbf{W}\mathbf{v} = \{\vec{v}'_1, \vec{v}'_2, \dots, \vec{v}'_{N_v}\}$, $\vec{v}'_i \in \mathbb{R}^{H'_c}$, where $\mathbf{W} \in \mathbb{R}^{H'_c \times H_c}$ is a weight matrix and H'_c is output dimensional of vertex hidden vector. The attention coefficient α_{ij} indicates the importance of the j th vertex features to the i th vertex, and it can be computed as follows:

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(\vec{a}^T [\vec{v}'_i || \vec{v}'_j]))}{\sum_{k \in \mathcal{N}_i} \exp(\text{LeakyReLU}(\vec{a}^T [\vec{v}'_i || \vec{v}'_k]))}, \quad (1)$$

where $\mathcal{N}_i$ is the neighbors of the i th vertex, $\vec{a}$ is a shared attention mechanism, parameterized by a weight vector $\vec{a} \in \mathbb{R}^{2H'_c}$, and $||$ is the concatenation operation. We then use the normalized attention

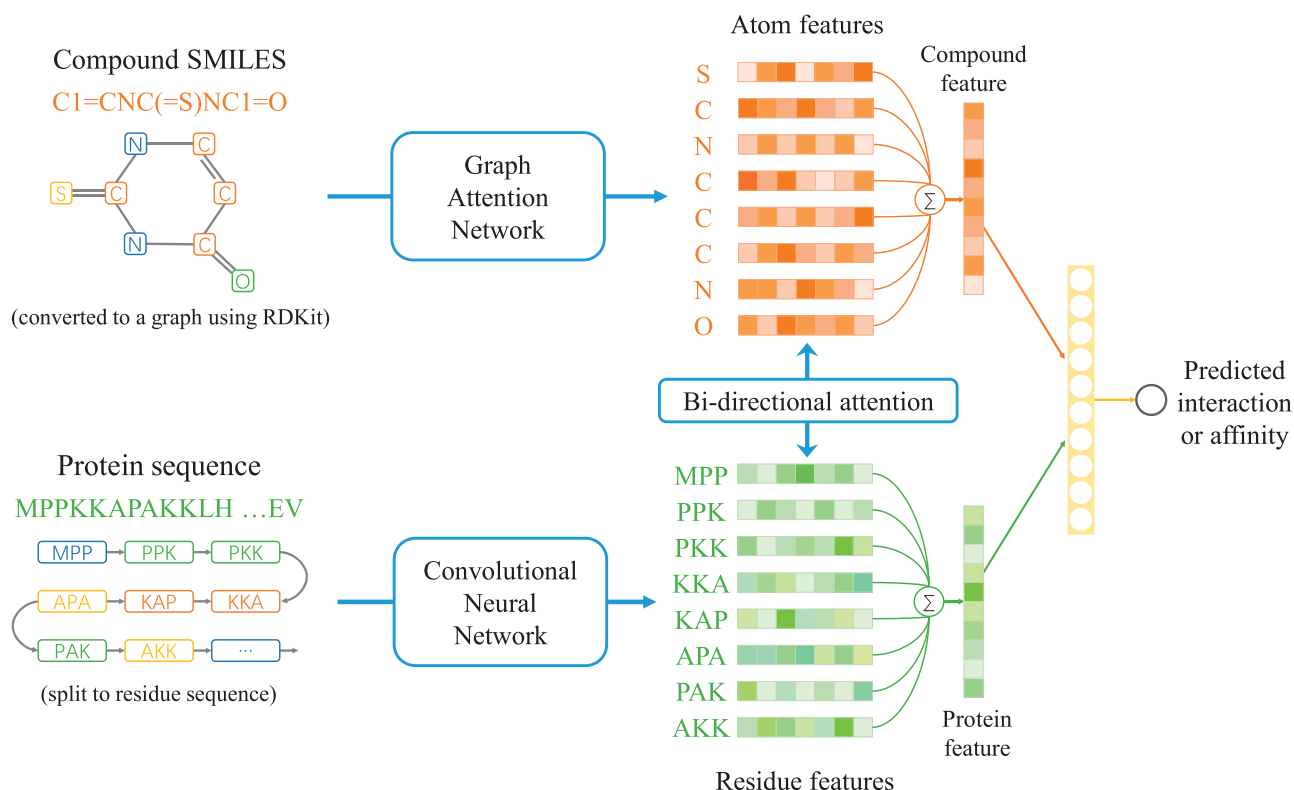


Fig. 1. The overview of BACPI. Given a compound–protein pair, a GAT and a CNN are used to learn the representations of the atom and residue from the compound structure graph and protein sequence, respectively. Then, these atom and residue representations are fed into the bi-directional attention neural network to integrate the representations and capture the important regions of compounds and proteins. Finally, the integrated features are used to predict the CPI or binding affinity

coefficients to update the vertex hidden vector which serves as the final output features for every vertex:

$$\vec{v}'_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} \vec{v}'_j \right). \quad (2)$$

In addition, we extend the mechanism to a multi-head attention by combining the output features of K independent graph attentional layer to stabilize the learning process of self-attention, that is,

$$\vec{v}_i^{cat} = \parallel \sigma \left(\sum_{j \in N_i} \alpha_{ij}^k \vec{v}_j^{rk} \right), \quad (3)$$

where $\parallel$ is the concatenation operation, $\vec{v}_i^{cat} \in \mathbb{R}^{KH_c}$ is the concatenation of K different layer, and α^k are the normalized attention coefficients of k th graph attentional layer. Finally, a single-layer neural network is used to transform $\{\vec{v}_i^{cat}\}_{i=1}^{N_r}$ into a compound space, that is,

$$\vec{v}_i = \text{LeakyReLU}(\mathbf{W}_{out} \vec{v}_i^{cat}), \quad (4)$$

where $\mathbf{W}_{out} \in \mathbb{R}^{H_c \times KH_c}$ is transition parameter, and $\vec{v}_i \in \mathbb{R}^{H_c}$ is one of the final atom features. While, this is the GAT module applied to obtain the vector representations of each atom from compound represented by graph.

In this study, we also use the extended connectivity fingerprints (ECFPs) (Rogers and Hahn, 2010) as the representation of the compound. ECFPs are a class of topological fingerprints for molecular substructure characterization, which describe the characteristics of substructures consisting of each atom and circular neighborhoods within a diameter range. We use RDKit to calculate the fingerprint of compounds and obtain a feature vector $\vec{f}$ of length 1024. The feature vector $\vec{f}$ is finally transformed into the compound space by multi-layer neural network.

2.2 Protein representation learning

Recently, CNN has been widely used for processing sequence-based data (e.g. protein sequences, text sequences and image sequences). As shown in Figure 1, protein is represented as amino acid sequences, and CNN is used to extract the local contextual features of protein and learn vector representations of protein. Specifically, to capture diverse subsequences patterns, we introduce a context window w to split the protein sequences into overlapping subsequences of amino acids. Since there are 20 types of amino acids, the total possible subsequences of amino acids is 20^w . A larger context window w enables the model to capture more subsequences patterns, which can effectively improve the prediction performance, and meanwhile it requires more computational resources and training data. In this study, to balance the prediction performance of the model and the consumption of computational resource, we set the context window $w=3$. For example, protein sequence *MRPSG...FIGA* can be split into diverse subsequences: ‘MRP’, ‘RPS’, ‘PSG’,..., ‘FIG’, ‘IGA’ of a fixed length 3. In this article, these fixed-length subsequences are regarded as amino acid residues. Let $S = \{s_1, s_2, \dots, s_{L_p}\}$ be a protein amino acid sequence, where s_i is the i th amino acid and L_p is the length of the protein sequence. Firstly, we split the protein sequence S into a residue sequence $R = \{r_1, r_2, \dots, r_{N_r}\}$, where r_i is the i th residue and N_r is the number of residues and equals to $L_p - w + 1$. We then translate all residues to randomly initialized embeddings $\mathbf{R} = \{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_{N_r}\}$, $\vec{r}_i \in \mathbb{R}^{H_p}$, where H_p is the dimension of residue embeddings. And the randomly initialized embeddings are updated through several convolutional layers with a non-linear activation function (e.g. ReLU). The hyperparameters of CNN mainly include the number of convolutional layers, the size and the number of filters. Finally, we obtain the final output $\{\vec{r}_i \in \mathbb{R}^{H_p}\}_{i=1}^{N_r}$ for all residues along the protein sequence.

2.3 CPI and affinity prediction

Neural networks with attention mechanism have been widely used to enhance the performance in various tasks (e.g. natural language processing and recommendation system), which can capture the importance of different input positions based on attention weights. In this work, we develop a bi-directional attention neural network that can produce attentions in both the directions, atom-to-residue and residue-to-atom, for compound and protein (each with a list of features), and thus enables the input compound-protein pairs to be aware of each other.

First, the atom features $\{\vec{v}_i\}_{i=1}^{N_v}$, the fingerprint features $\vec{f}$ and the residue features $\{\vec{r}_i\}_{i=1}^{N_r}$, are transformed into vectors with same dimension d by single-layer neural network, that is,

$$\vec{c}_i = \text{LeakyReLU}(\mathbf{W}_v \vec{v}_i), \quad (5)$$

$$\vec{h}_f = \text{LeakyReLU}(\mathbf{W}_f \vec{f}), \quad (6)$$

$$\vec{p}_i = \text{LeakyReLU}(\mathbf{W}_r \vec{r}_i), \quad (7)$$

where $\mathbf{W}_v \in \mathbb{R}^{d \times H_c}$, $\mathbf{W}_f \in \mathbb{R}^{d \times H_c}$, $\mathbf{W}_r \in \mathbb{R}^{d \times H_p}$, d is the unified dimension in bi-directional attention neural network module, and $\{\vec{c}_i\}_{i=1}^{N_v}$, $\vec{h}_f$, and $\{\vec{p}_i\}_{i=1}^{N_r}$ are the vectors of transformed atom features, fingerprint features, and residue features, respectively. Suppose $\mathbf{C} = [\vec{c}_1^T, \vec{c}_2^T, \dots, \vec{c}_{N_v}^T] \in \mathbb{R}^{N_v \times d}$ is the context matrix of a given compound, as well as $\mathbf{P} = [\vec{p}_1^T, \vec{p}_2^T, \dots, \vec{p}_{N_r}^T] \in \mathbb{R}^{N_r \times d}$ is the context matrix of a given protein. A soft alignment matrix $\mathbf{A} \in \mathbb{R}^{N_v \times N_r}$ is calculated as

$$\mathbf{A} = \tanh(\mathbf{C} \mathbf{U} \mathbf{P}^T) \quad (8)$$

where $\mathbf{U} \in \mathbb{R}^{d \times d}$ is a trainable parameter matrix, and $\mathbf{A}$ is a pairwise interaction matrix and $\mathbf{A}_{ij}$ can be assumed to represent the interaction strength between i th atom and j th residue. We extract the information weighted by $\mathbf{A}$ for each atom of the compound from all the residues of the protein, and vice versa for each residue of the protein, that is,

$$\mathbf{I}_c = \mathbf{A} \tanh(\mathbf{P} \mathbf{W}_{r2a}), \quad (9)$$

$$\mathbf{I}_p = \mathbf{A}^T \tanh(\mathbf{C} \mathbf{W}_{a2r}), \quad (10)$$

where $\mathbf{W}_{r2a}, \mathbf{W}_{a2r} \in \mathbb{R}^{d \times d}$, $\mathbf{I}_c \in \mathbb{R}^{N_v \times d}$ are the information delivered from residues to atoms, and $\mathbf{I}_p \in \mathbb{R}^{N_r \times d}$ are the information delivered from atoms to residues. Next, the attentions in both directions, atom-to-residue α_{a2r} and residue-to-atom α_{r2a} , are calculated as follows:

$$\alpha_{a2r} = \text{softmax}([\mathbf{C} \mathbf{W}_c || \mathbf{I}_c] \mathbf{a}_{a2r}), \quad (11)$$

$$\alpha_{r2a} = \text{softmax}([\mathbf{P} \mathbf{W}_p || \mathbf{I}_p] \mathbf{a}_{r2a}), \quad (12)$$

where $\mathbf{W}_c, \mathbf{W}_p \in \mathbb{R}^{d \times d}$, $||$ is the concatenation operation, $\mathbf{a}_{a2r} \in \mathbb{R}^{2d}$ and $\mathbf{a}_{r2a} \in \mathbb{R}^{2d}$ are the attention mechanisms of atom-to-residue and residue-to-atom, $\alpha_{a2r} \in \mathbb{R}^{N_v}$ and $\alpha_{r2a} \in \mathbb{R}^{N_r}$ are the compound and protein attentions normalized by the softmax function. Finally, the compound feature $\vec{h}_c$ and protein feature $\vec{h}_p$ are the weighted sums of atom features and residue features based on the normalized attentions α_{a2r} and α_{r2a} respectively, that is,

$$\vec{h}_c = \alpha_{a2r} \cdot \mathbf{C}, \quad (13)$$

$$\vec{h}_p = \alpha_{r2a} \cdot \mathbf{P}, \quad (14)$$

where $\vec{h}_c, \vec{h}_p \in \mathbb{R}^d$.

To stabilize the learning process of the bi-directional attention, we extend the attention mechanism to multi-head attention. Specifically, L independent bi-directional attention mechanisms execute the Equation (8–14) to obtain different compound features and

protein features, and then the different compound features and protein features are concatenated. Fully expanded out, the concatenated compound features $\vec{h}_c^{cat}$ and protein features $\vec{h}_p^{cat}$ can be expressed as:

$$\vec{h}_c^{cat} = \big\|_{l=1}^L \text{softmax}([\mathbf{C} \mathbf{W}_c^l || \mathbf{A}^l \tanh(\mathbf{P} \mathbf{W}_{r2a}^l)] \mathbf{a}_{a2r}^l) \cdot \mathbf{C} \quad (15)$$

$$\vec{h}_p^{cat} = \big\|_{l=1}^L \text{softmax}([\mathbf{P} \mathbf{W}_p^l || \mathbf{A}^l \tanh(\mathbf{C} \mathbf{W}_{a2r}^l)] \mathbf{a}_{r2a}^l) \cdot \mathbf{P} \quad (16)$$

$$\mathbf{A}^l = \tanh(\mathbf{C} \mathbf{U}^l \mathbf{P}^T), \quad (17)$$

where $\vec{h}_c^{cat}, \vec{h}_p^{cat} \in \mathbb{R}^{Ld}$. The concatenated features $\vec{h}_c^{cat}$ and $\vec{h}_p^{cat}$ are then transformed into dimension d by single-layer neural network, resulting in the following final feature representation of compound and protein:

$$\vec{h}_c^{final} = \mathbf{W}_{fc} \vec{h}_c^{cat}, \quad (18)$$

$$\vec{h}_p^{final} = \mathbf{W}_{fp} \vec{h}_p^{cat}, \quad (19)$$

where $\mathbf{W}_{fc}, \mathbf{W}_{fp} \in \mathbb{R}^{d \times Ld}$ and $\vec{h}_c^{final}, \vec{h}_p^{final} \in \mathbb{R}^d$. Finally, the CPI or binding affinity is predicted as follows:

$$y = \mathbf{W}_y \text{LeakyReLU}(\text{flatten}([\vec{h}_c^{final} || \vec{h}_p^{final}] \otimes \vec{h}_p^{final})), \quad (20)$$

where the concatenation of compound feature $\vec{h}_c^{final}$ and fingerprint feature $\vec{h}_f$ is the representation of the overall compound feature, $\otimes$ denotes the outer product, and the outer product of compound and protein representation is flattened into a column vector of length $2d^2$. Note that, in CPI prediction task, the output result contains two values representing the probability of interaction and non-interaction predicted by a binary classifier, in which $\mathbf{W}_y \in \mathbb{R}^{2 \times 2d^2}$. And in compound-protein binding affinity prediction task, the output result is a continuous value representing the binding affinity predicted by a linear regression, in which $\mathbf{W}_y \in \mathbb{R}^{1 \times 2d^2}$.

3 Experiments and results

3.1 Datasets

In binary classification-based CPI prediction studies, compound-protein pairs are labeled as positive or negative samples. The selection of negative samples is a well-known problem. Most of the supervised learning methods treat the known experimentally validated CPIs as positive samples and the unknown CPIs or randomly select a part of unconfirmed CPIs as negative samples. However, such random negative samples may include real positive samples not yet known and would largely influence the predictive accuracy and credibility of the model. To solve the problem, screening true negative samples is therefore necessary to construct highly credible CPI datasets (Ding et al., 2014). In this work, we use human and *C.elegans* CPI datasets created by Liu et al. (2015) to evaluate the classification performance of our model in CPI prediction task. The positive samples of the datasets were collected from DrugBank 4.1 (Wishart et al., 2008), Matador (Günther et al., 2008) and STITCH 4.0 (Kuhn et al., 2014), which contains 3369 positive interactions for human datasets and 4000 positive interactions for *C.elegans* datasets. The highly credible negative samples were retrieved by a systematic screening framework integrated various resources of compounds and proteins, and totally obtained 384 916 negative samples for human datasets and 88 261 negative samples for *C.elegans* datasets.

In regression-based compound-protein binding affinity prediction studies, compound-protein pairs are labeled as binding affinity values from BindingDB (Liu et al., 2007). There are four measures of binding affinity: IC_{50} , K_i , K_d and EC_{50} . In this work, we use four

Table 1. The number of distinct compounds, proteins, train and test samples of IC₅₀, EC₅₀, K_i and K_d datasets

Dataset	No. of compound	No. of protein	No. of train	No. of test
IC ₅₀	265 627	2795	263 583	113 168
EC ₅₀	31 970	513	26 413	11 483
K _i	93 437	1619	101 134	43 391
K _d	5895	812	8778	3811

Note: These datasets were created by Karimi *et al.* (2019) from BindingDB.

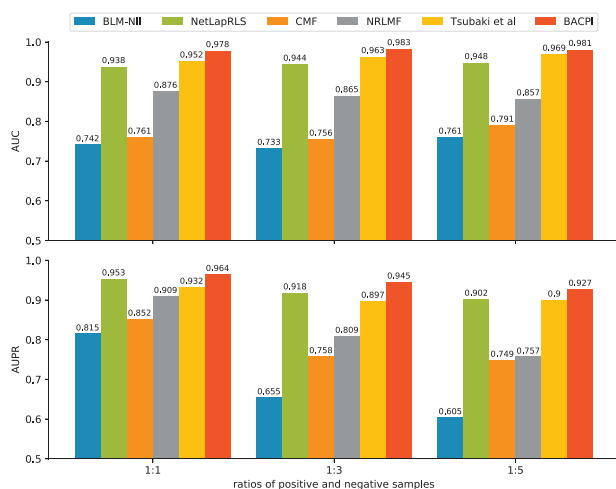


Fig. 2. The AUC and AUPR values of BLM-NII, NetLapRLS, CMF, NRLMF, Tsubaki *et al.* (2019) and the proposed BACPI method on human datasets with different ratios of positive and negative samples for CPIs prediction

datasets created by Karimi *et al.* (2019) to evaluate the regression performance of our model in the compound–protein binding affinity prediction task. These datasets contain 489 280 samples for IC₅₀, 144 525 samples for K_i, 12 589 samples for K_d and 37 896 samples for EC₅₀, respectively. We use the measure of binding affinity in their logarithm form (i.e. $\text{pIC}_{50} = -\log_{10}\text{IC}_{50}$) as the target label for regression. Table 1 shows basic statistics of four datasets. Each dataset is split into the training set and the test set, and the training set including 10% held out for validation.

3.2 Implementation and training

Our method uses features obtained by end-to-end learning of GAT and CNN for representations of compounds and proteins. For compound, we use RDKit to convert the SMILES format of the compound into a graph representation and employ GAT to learn the feature representation of the compound which can extract various information of the graph, such as atom types, aromaticity and chemical bond types. For protein, our CNN takes the amino acid sequence of the protein as input and learn the feature representation of the protein. Finally, our bi-directional attention neural network is used to integrate the representations of compounds and proteins and predict the interaction and binding affinity of input compound–protein pair. Given a set of compound–protein pairs and the labels (interaction or affinity), the training objective is to minimize the loss function (Cross-Entropy Loss in CPI prediction task and Mean Square Error in affinity prediction task) and use backpropagation to optimize all weight matrices and bias vectors in our GAT, CNN and bi-directional attention neural network. The optimizer of our model is Adam (Kingma and Ba, 2014).

3.3 CPI prediction

Following previous studies (Ding *et al.*, 2014; Liu *et al.*, 2016; Tsubaki *et al.*, 2019), we examined the CPI prediction performance of BACPI method evaluated via five times repeated 5-fold cross-

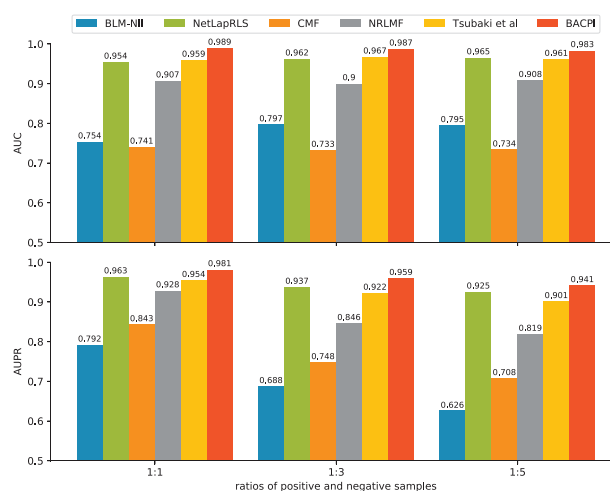


Fig. 3. The AUC and AUPR values of BLM-NII, NetLapRLS, CMF, NRLMF, Tsubaki *et al.* (2019) and the proposed BACPI method on *C.elegans* datasets with different ratios of positive and negative samples for CPIs prediction

validation (5×5 -fold CV). In each of the five repetitions, the CPIs in datasets are randomly assigned into five equal size splits, each split is used as a test set and the remaining four splits are used as training set (including 10% held out for validation) of a predictive model. We adopt area under the receiver operating characteristic curve (AUC) and area under precision-recall curve (AUPR) as evaluation metrics and conduct performance evaluation on the human and *C.elegans* datasets from Liu *et al.* (2015). Since real-world CPI datasets are typically unbalanced, we set different ratios of positive and negative samples (e.g. 1:1, 1:3 and 1:5) to evaluate the robustness of the predictive model. The proposed BACPI method was compared with five competing methods: BLM-NII (Mei *et al.*, 2013), NetLapRLS (Xia *et al.*, 2010), CMF (Zheng *et al.*, 2013), NRLMF (Liu *et al.*, 2016) and an end-to-end method (Tsubaki *et al.*, 2019). Note that the four methods of BLM-NII, NetLapRLS, CMF and NRLMF take a CPI matrix, chemical structure similarity matrix and protein sequence similarity matrix as inputs. We computed the chemical structure similarity between compounds and sequence similarity between proteins for BLM-NII, NetLapRLS, CMF and NRLMF, which were obtained by using the Jaccard score (Jaccard, 1908) between Morgan fingerprints calculated by RDKit and the normalized of Smith–Waterman score (Smith *et al.*, 1981), respectively.

Figures 2 and 3 show the AUC and AUPR scores on the human and *C.elegans* datasets. As can be seen, on both datasets, the proposed BACPI method achieved the best performance compared to these methods, and the method proposed by Tsubaki *et al.* (2019) also achieved competitive performance. This suggests that the end-to-end representation learning method can explore useful compound and protein properties for downstream prediction. We also found that when the ratio of positive and negative samples increases, the AUC scores of most methods keep steady or increase slightly, while the AUPR scores of all methods decrease. This is mainly due to both recall and precision are focused on the positive class (the minority class), which leads to AUPR punishes false-positives more than AUC (Davis and Goadrich, 2006). Therefore, when dealing with an unbalanced dataset, AUPR gives a more accurate evaluation of a method's performance, while AUC may provide an optimistic view of the performance. As one can see, on the unbalanced datasets (the ratio of positive and negative samples is 1:3 or 1:5), our method significantly outperformed other methods on both human and *C.elegans* datasets in terms of AUPR. These experiment results demonstrate that BACPI is reliable and robust even on unbalanced datasets.

3.4 Compound–protein binding affinity prediction

To evaluate the performance of our model in compound–protein binding affinity prediction, we examined BACPI method on four datasets of different affinity measures (i.e. IC₅₀, EC₅₀, K_i, K_d) from

Table 2. The RMSE scores of different prediction approaches on IC50, EC50, K_i and K_d datasets

Method	IC50	EC50	K_i	K_d
Ridge	1.23	0.93	1.27	1.24
Lasso	1.22	0.98	1.27	1.28
RF	0.91	0.80	0.97	1.10
DeepAffinity	0.86	0.85	0.96	1.17
MONN	0.82	0.79	0.88	1.07
DeepPurpose	0.85	0.80	0.92	1.16
BACPI	0.74	0.78	0.80	1.08

Note: Bold numbers represent the best performance of each dataset.

Table 3. The PCC scores of different prediction approaches on IC50, EC50, K_i and K_d datasets

Method	IC50	EC50	K_i	K_d
Ridge	0.54	0.77	0.58	0.60
Lasso	0.55	0.74	0.58	0.55
RF	0.78	0.84	0.78	0.70
DeepAffinity	0.82	0.83	0.79	0.68
MONN	0.83	0.85	0.83	0.74
DeepPurpose	0.83	0.85	0.81	0.69
BACPI	0.86	0.85	0.86	0.73

Note: Bold numbers represent the best performance of each dataset.

Karimi et al. (2019) and compare it with the shallow machine learning models: ridge regression, lasso regression and random forest (RF) and the deep learning models: DeepAffinity (Karimi et al., 2019), MONN (Li et al., 2020) and DeepPurpose (Huang et al., 2021). Note that, the results of ridge regression, lasso regression and RF are obtained in Karimi et al. (2019). For DeepAffinity, MONN and DeepPurpose, we used their default setting to evaluate the performance. Table 2 shows the root mean squared error (RMSE) results of various traditional machine learning methods and the deep learning methods. As can be seen, the deep learning methods achieved higher accuracy compared to machine learning methods: ridge regression, lasso regression and RF. This is mainly due to the powerful feature learning ability of deep learning. MONN, DeepPurpose and the proposed method outperformed DeepAffinity since DeepAffinity is not an end-to-end learning method, which need a large amount of unlabeled data for unsupervised pre-training to encode compound SMILES or protein SPS into representations. In addition, the proposed method achieved comparable performance on small datasets (i.e. EC50, K_d) and higher performance on large datasets (i.e. IC50, K_i) compared to MONN. MONN relies on the information of local non-compared interactions between compounds and proteins, it is difficult to accurately capture the local binding sites without extra supervised guidance. The proposed method can automatically capture the important regions of compounds and proteins through a multi-head bi-directional attention mechanism, especially when the training data sample is sufficient.

Table 3 shows the Pearson correlation coefficient (PCC) results of different prediction approaches on four datasets. BACPI method achieved the best results on IC50 and K_i datasets, the same result with MONN and DeepPurpose on EC50 dataset and second best on K_d dataset. All the comparison results indicate that BACPI can be effectively applied to predict the binding affinity of compounds and proteins.

3.5 Case study

To test the effectiveness of BACPI in drug repurposing, we predicted the binding affinity between existing drugs and SARS-CoV2 3C-like Protease (3CL^{Pro}) target. The existing drugs include 82 antiviral drugs (Huang et al., 2021), 1 antiparasitic drug (Ivermectin), 1 anti-malarial drug (Artemisinin) and 3 unrelated drugs (Amoxicillin,

Table 4. Prediction results of the binding affinity between existing drugs and SARS-CoV2 3C-like protease

Rank	Drug	Affinity	Source
1	Darunavir	2.16	Costanzo et al. (2020)
2	Simeprevir	9.75	Lo et al. (2021)
3	Cobicistat	10.0	Chen et al. (2020)
4	Grazoprevir	13.03	
5	Ritonavir	14.58	Costanzo et al. (2020)
6	Glecaprevir	15.87	Shamsi et al. (2020)
7	Enfuvirtide	17.56	
8	Tipranavir	23.08	Dayer (2020)
9	Ivermectin	27.14	López-Medina et al. (2021)
10	Amprenavir	46.22	
78	Amoxicillin	18 817.04	
82	Penicillin	22 743.94	
84	Aspirin	34 734.3	

Note: The prediction score of binding affinity is measured in K_i [μ M]. The smaller the K_i value, the greater the binding affinity.

Penicillin, and Aspirin). The 3CL^{Pro} target is the main protease found in coronaviruses and plays an essential role in viral replication. We obtained the SMILES of these 87 existing drugs and the amino acid sequences of the 3CL^{Pro} target from the PubChem (Kim et al., 2016) database and the Protein Data Bank (Berman et al., 2000) database. The pre-trained BACPI takes these SMILES and amino acid sequences as input, and outputs top-ranked drug candidates. Furthermore, to ensure that the top-ranked drug candidates are not predicted by chance, we aggregated 100 pre-trained models to obtain stable prediction results. The prediction results of the top-10 drug candidates and 3 unrelated drugs are shown in Table 4.

These results show that 7 recommendations out of 10 have been demonstrated their replication inhibitory effect against SARS-CoV2 by many studies. And we find that Darunavir, Cobicistat, Ritonavir and Ivermectin are four of drug candidates that are currently undergo clinical trials for treating coronavirus disease 2019 (Chen et al., 2020; Costanzo et al., 2020; López-Medina et al., 2021). In contrast, three unrelated drugs (Amoxicillin, Penicillin and Aspirin) showed weak attraction and binding affinity with the 3CL^{Pro} target, ranking 78, 80 and 82 out of 87, respectively. These experimental results confirm the reliability of BACPI in screening repurposing candidates.

4 Conclusion

The identification of CPIs is an essential step in the process of drug discovery. However, the experimental determination of CPIs is still expensive and time consuming, although various high-throughput systems biology assays are available. In structure-based prediction, approaches are known for the heavy dependence on the available 3D-structure data. Therefore, several machine learning-based and deep learning-based methods have been proposed to predict CPIs or binding affinity. In this article, we propose an end-to-end representation learning approach to predict CPIs and binding affinity using only molecule graphs of compounds and sequences of proteins. We use GAT and CNN to learn representations from raw data of compounds and proteins, and bi-directional attention neural network in CPI prediction task or the affinity prediction task. Comparison tests showed that BACPI significantly outperformed other methods on both human and *C.elegans* datasets in predicting CPIs. And in the binding affinity prediction task, BACPI achieved the best performance on large datasets (i.e. IC50, K_i) or comparable performance on small datasets (i.e. EC50, K_d) compared to MONN. Based on these evaluations, we can conclude that BACPI can be effectively applied to predict the interaction and the binding affinity of compounds and protein, and thus greatly advancing the process of drug development.

It is also worth noting that BACPI can automatically capture the important regions of compounds and proteins by highlighting high-value attention weights. And such important regions with high weight values were computed through a multi-head bi-directional attention mechanism without extra supervised guidance. We observed that BACPI performs better on large datasets compared to the performance on small datasets. This might be an indication that the bi-directional attention neural networks require a large amount of training data to learn accurate attention weights of compounds and proteins. As future work, we will construct a benchmark dataset containing massive amounts of data from several databases, such as BindingDB (Liu *et al.*, 2007), DUD-E (Mysinger *et al.*, 2012), Protein Data Bank (Berman *et al.*, 2000) and PDBbind (Wang *et al.*, 2005). We believe that our study will facilitate virtual high-throughput screening and accurately predict the binding sites between compounds and proteins with massive data.

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